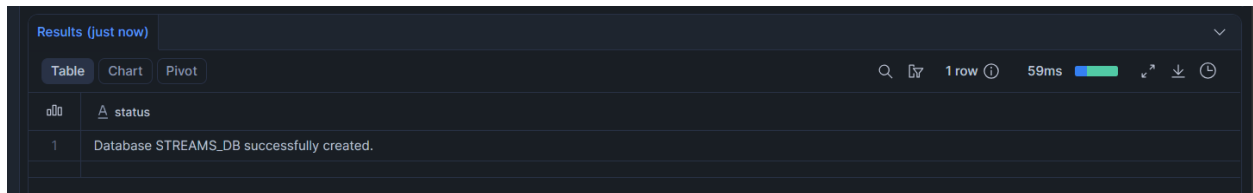


ASSIGNMENT – 2

TASK 1: Create Database

```
CREATE OR REPLACE TRANSIENT DATABASE STREAMS_DB;
```

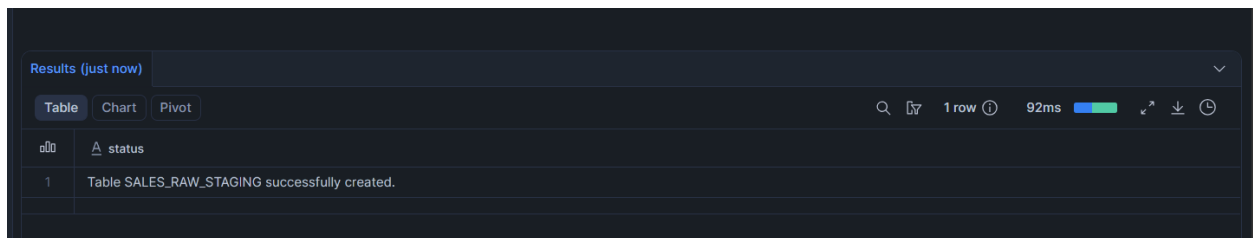


The screenshot shows a database query results window with a dark theme. At the top, it says "Results (just now)". Below that are tabs for "Table", "Chart", and "Pivot". To the right of the tabs are icons for search, filters, and a status bar indicating "1 row" and "59ms". The main area displays a single row with a status icon and the text "Database STREAMS_DB successfully created."

	status
1	Database STREAMS_DB successfully created.

TASK 2: Create Raw Sales Staging Table

```
CREATE OR REPLACE TABLE SALES_RAW_STAGING(  
    id int,  
    product string,  
    price float,  
    amount int,  
    store_id int  
);
```



The screenshot shows a database query results window with a dark theme. At the top, it says "Results (just now)". Below that are tabs for "Table", "Chart", and "Pivot". To the right of the tabs are icons for search, filters, and a status bar indicating "1 row" and "92ms". The main area displays a single row with a status icon and the text "Table SALES_RAW_STAGING successfully created."

	status
1	Table SALES_RAW_STAGING successfully created.

TASK 3: Insert Initial Sales Data

```
INSERT INTO SALES_RAW_STAGING VALUES
```

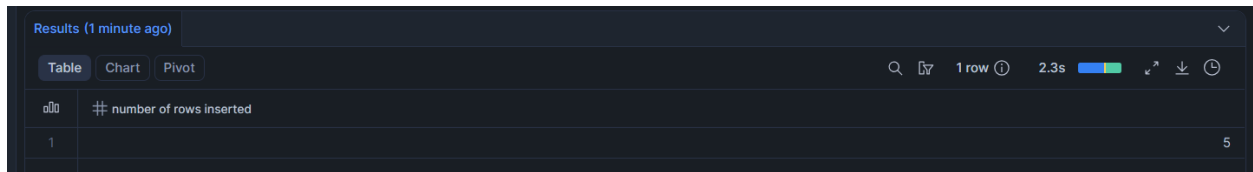
```
(1,'Banana',1.99,1,1),
```

```
(2,'Lemon',0.99,1,1),
```

```
(3,'Apple',1.79,1,2),
```

```
(4,'Orange Juice',1.89,1,NULL),
```

```
(5,'Cereals',5.98,2,1);
```



Results (1 minute ago)

Table Chart Pivot

1 row 2.3s

	# number of rows inserted
1	5

TASK 4: Create Store Table

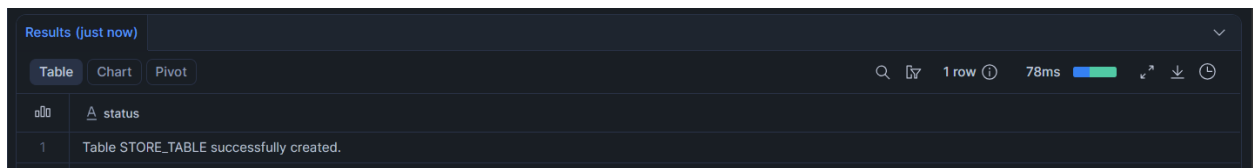
```
CREATE OR REPLACE TABLE STORE_TABLE (
```

```
store_id int,
```

```
location string,
```

```
employees int
```

```
);
```



Results (just now)

Table Chart Pivot

1 row 78ms

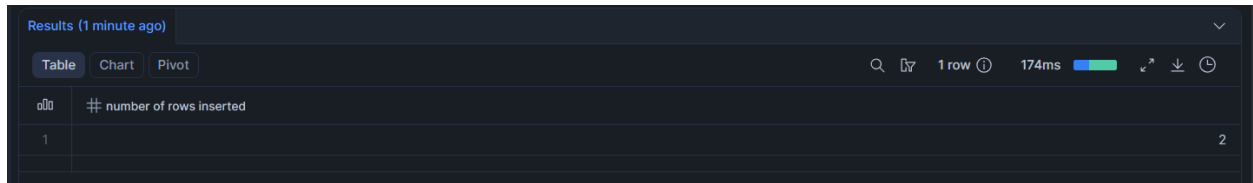
	status
1	Table STORE_TABLE successfully created.

TASK 5: Insert Store Data

```
INSERT INTO STORE_TABLE VALUES
```

```
(1, 'Chicago', 33),
```

```
(2, 'London', 12);
```



number of rows inserted	2
1	2

TASK 6: Create Final Sales Table

```
CREATE OR REPLACE TABLE SALES_FINAL_TABLE (
```

```
    id INT,
```

```
    product STRING,
```

```
    price FLOAT,
```

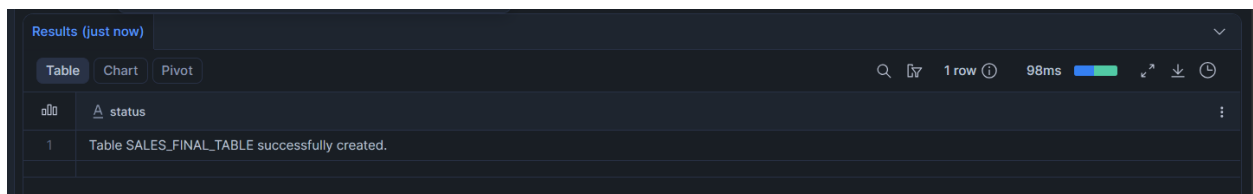
```
    amount INT,
```

```
    store_id INT,
```

```
    location STRING,
```

```
    employees INT
```

```
);
```



status	Table SALES_FINAL_TABLE successfully created.
1	Table SALES_FINAL_TABLE successfully created.

TASK 7: Perform Initial Full Load

```
INSERT INTO SALES_FINAL_TABLE

SELECT

    s.id, s.product,

    s.price,

    s.amount,

    s.store_id,

    st.location,

    st.employees

FROM SALES_RAW_STAGING s

LEFT JOIN STORE_TABLE st

ON s.store_id = st.store_id;
```

Results (just now)	
Table	Chart Pivot
1 row	282ms
# number of rows inserted	5

TASK 8: Verify Initial Load

```
SELECT * FROM SALES_RAW_STAGING;

SELECT * FROM STORE_TABLE;

SELECT * FROM SALES_FINAL_TABLE;
```

Results (2 minutes ago)

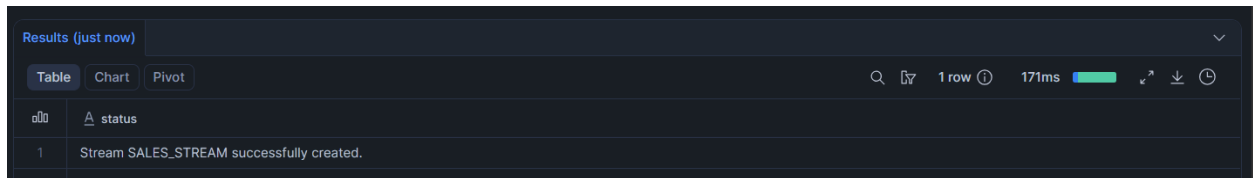
TableChartPivot

5 rows30ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Banana	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33

TASK 9: Stream Implementation (Create Stream Object):

```
CREATE OR REPLACE STREAM SALES_STREAM  
ON TABLE SALES_RAW_STAGING;
```



Results (just now)

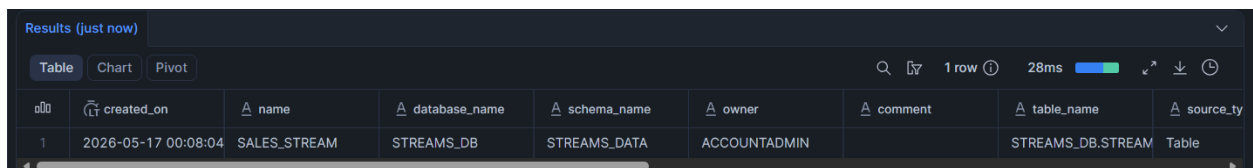
Table Chart Pivot

1 row 171ms

	status
1	Stream SALES_STREAM successfully created.

TASK 10: View Stream Information

```
SHOW STREAMS;  
DESC STREAM SALES_STREAM;  
SELECT * FROM SALES_STREAM;
```



Results (just now)

Table Chart Pivot

1 row 28ms

	created_on	name	database_name	schema_name	owner	comment	table_name	source_ty
1	2026-05-17 00:08:04	SALES_STREAM	STREAMS_DB	STREAMS_DATA	ACCOUNTADMIN		STREAMS_DB.STREAM	Table

QUESTION: Why is SALES_STREAM empty immediately after creation?

ANSWER: SALES_STREAM is empty immediately after creation because streams only capture changes made AFTER the stream is created.

TASK 11: Insert New Sales Records

```
INSERT INTO SALES_RAW_STAGING VALUES  
(6, 'Mango', 1.99, 1, 2),  
(7, 'Garlic', 0.99, 1, 1);
```

Results (just now)

Table Chart Pivot

1 row 2.2s

#	number of rows inserted
1	2

TASK 12: Check Stream Data

```
SELECT * FROM SALES_STREAM;
```

QUESTION: What value do you see in METADATA\$ACTION for newly inserted records?

ANSWER: FOR NEWLY INSERTED RECORDS

METADATA\$ACTION = 'INSERT'

Results (2 minutes ago)

Table Chart Pivot

2 rows 166ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
1	6	Mango	1.99	1	2	INSERT	FALSE	60f67aa70da8c0ee495383b671cff7ea13
2	7	Garlic	0.99	1	1	INSERT	FALSE	7a188d5a1071b8fe08125ab9c5e447d5t

TASK 13: Verify Source and Target Tables

```
SELECT * FROM SALES_RAW_STAGING;
```

```
SELECT * FROM SALES_FINAL_TABLE;
```

QUESTION: Are the new records available in SALES_FINAL_TABLE before consuming the stream?

ANSWER: No.

The new records are NOT available in SALES_FINAL_TABLE before consuming the stream.

Results (1 minute ago)

Table Chart Pivot

5 rows 100ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Banana	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33

TASK 14: Consume Stream Data

```
INSERT INTO SALES_FINAL_TABLE

SELECT

    s.id,

    s.product,

    s.price,

    s.amount,

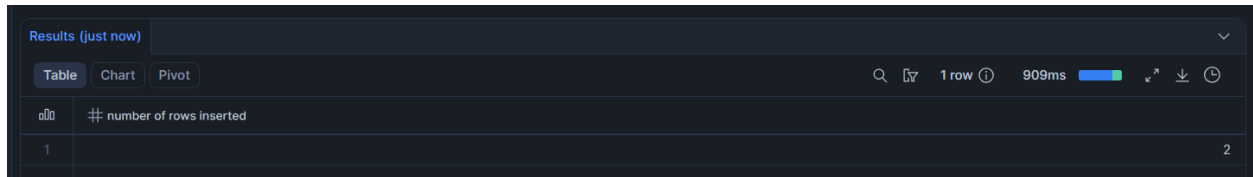
    s.store_id,

    st.location,

    st.employees

FROM SALES_STREAM s

LEFT JOIN STORE_TABLE st
```



The screenshot shows a database query results window with the title "Results (just now)". It has tabs for "Table", "Chart", and "Pivot". The "Table" tab is selected. The results show 1 row and 909ms execution time. The table has two columns: "number of rows inserted" and an unnamed column.

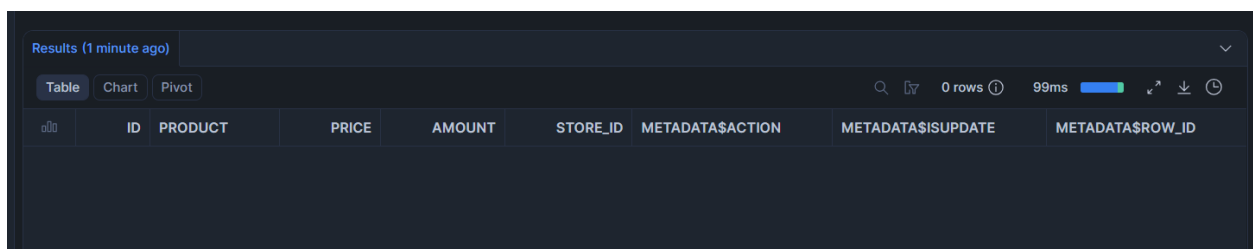
number of rows inserted	
1	2

TASK 15: Verify Stream After Consumption

```
SELECT * FROM SALES_STREAM;
```

QUESTION: Why did the stream become empty after inserting data into SALES_FINAL_TABLE?

ANSWER : The stream becomes empty because the change data was already consumed by the INSERT statement.

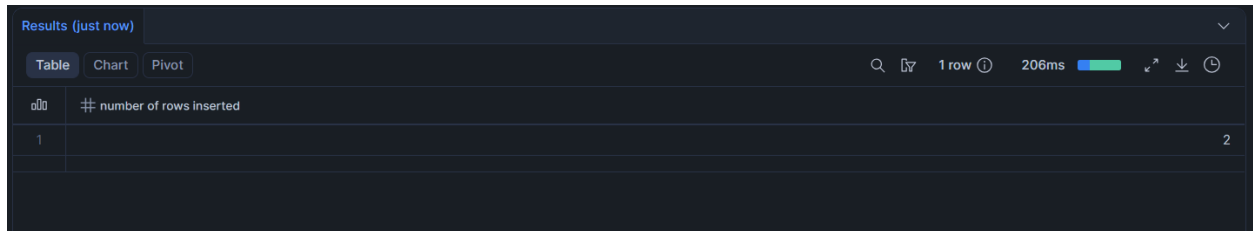


The screenshot shows a database query results window with the title "Results (1 minute ago)". It has tabs for "Table", "Chart", and "Pivot". The "Table" tab is selected. The results show 0 rows and 99ms execution time. The table has columns: ID, PRODUCT, PRICE, AMOUNT, STORE_ID, METADATA\$ACTION, METADATA\$ISUPDATE, and METADATA\$ROW_ID.

ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
----	---------	-------	--------	----------	------------------	--------------------	------------------

TASK 16: Insert More Sales Records

```
INSERT INTO SALES_RAW_STAGING VALUES  
  
(8, 'Paprika', 4.99, 1, 2),  
  
(9, 'Tomato', 3.99, 1, 2);
```

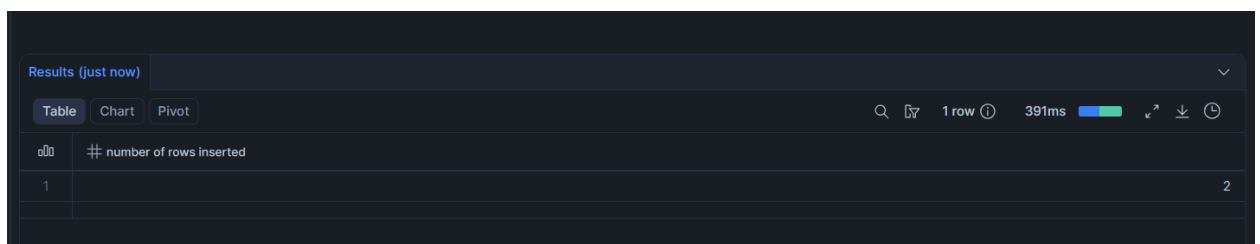


The screenshot shows a database query results window with a dark theme. The title bar says "Results (just now)". Below the title bar are tabs for "Table", "Chart", and "Pivot", with "Table" selected. To the right of the tabs are icons for search, filter, and a status bar showing "1 row", "206ms", and a progress bar. The main area displays a table with one row. The first column is labeled "id" and the second column is labeled "number of rows inserted". The row contains the values "1" and "2" respectively.

id	number of rows inserted
1	2

TASK 17 : Consume Second Stream Data

```
INSERT INTO SALES_FINAL_TABLE  
  
SELECT  
  
    s.id,  
  
    s.product,  
  
    s.price,  
  
    s.amount,  
  
    s.store_id,  
  
    st.location,  
  
    st.employees  
  
FROM SALES_STREAM s  
  
LEFT JOIN STORE_TABLE st  
  
ON s.store_id = st.store_id;
```



The screenshot shows a database query results window with a dark theme. The title bar says "Results (just now)". Below the title bar are tabs for "Table", "Chart", and "Pivot", with "Table" selected. To the right of the tabs are icons for search, filter, and a status bar showing "1 row", "391ms", and a progress bar. The main area displays a table with one row. The first column is labeled "id" and the second column is labeled "number of rows inserted". The row contains the values "1" and "2" respectively.

id	number of rows inserted
1	2

TASK 18 : Verify Final Insert Output

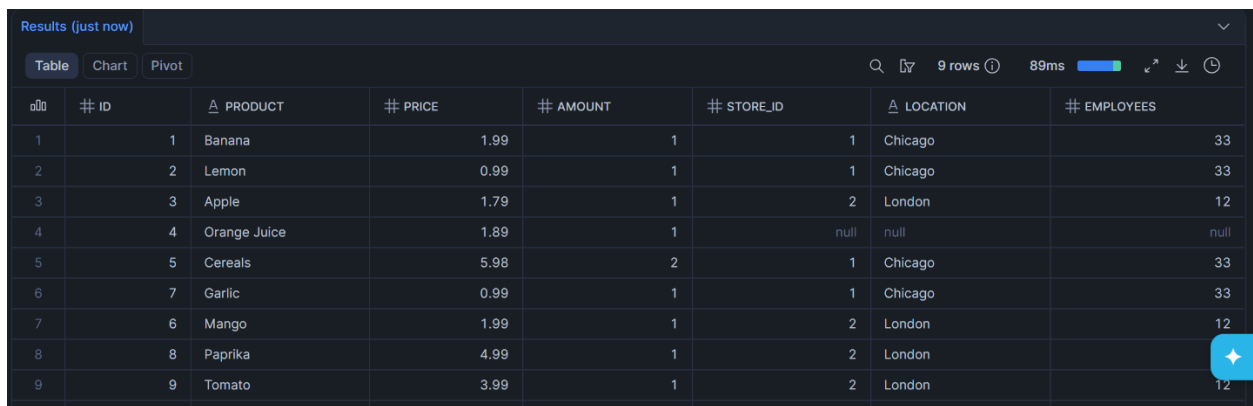
```
SELECT * FROM SALES_FINAL_TABLE;
```

```
SELECT * FROM SALES_RAW_STAGING;
```

```
SELECT * FROM SALES_STREAM;
```

QUESTION : How many records are now available in SALES_FINAL_TABLE?

ANSWER : SALES_FINAL_TABLE now contains **9 records**.



Results (just now)

Table Chart Pivot

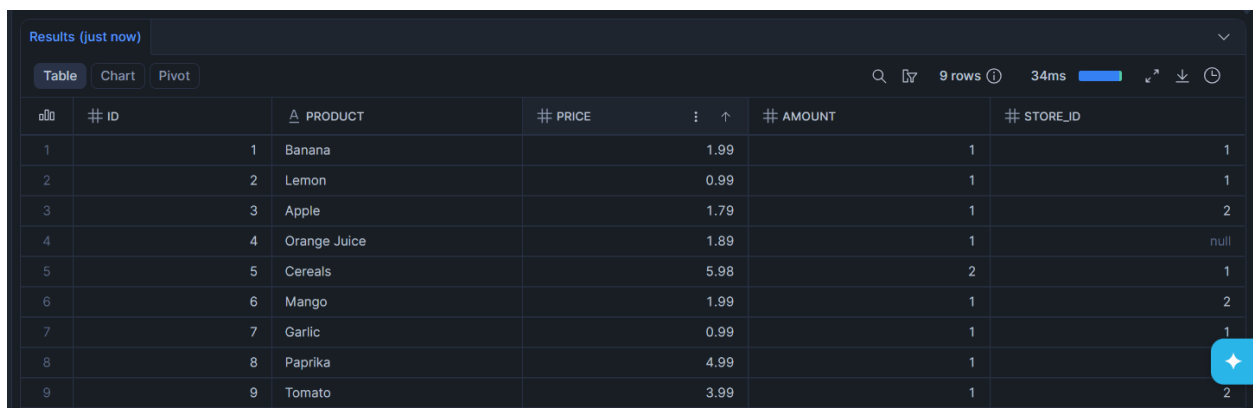
9 rows 89ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Banana	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	12
9	9	Tomato	3.99	1	2	London	12

TASK 19 : Check Existing Data Before Update

```
SELECT * FROM SALES_RAW_STAGING;
```

```
SELECT * FROM SALES_STREAM;
```

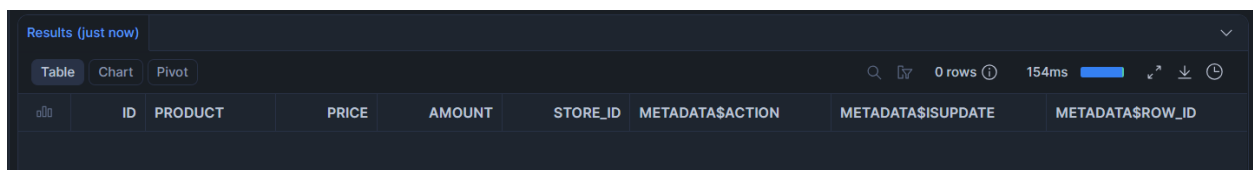


Results (just now)

Table Chart Pivot

9 rows 34ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Banana	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1
8	8	Paprika	4.99	1	1
9	9	Tomato	3.99	1	2



Results (just now)

Table Chart Pivot

0 rows 154ms

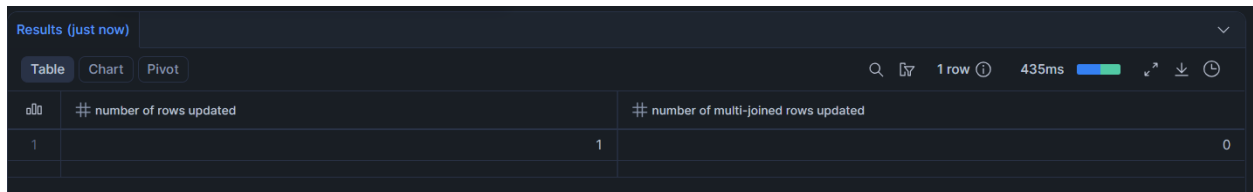
ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
----	---------	-------	--------	----------	------------------	--------------------	------------------

TASK 20 : Perform First Update

```
UPDATE SALES_RAW_STAGING
```

```
SET product = 'Potato'
```

```
WHERE product = 'Banana';
```



Results (just now)

Table Chart Pivot

1 row 435ms

	# number of rows updated	# number of multi-joined rows updated
1	1	0

TASK 21 : Check Stream After Update

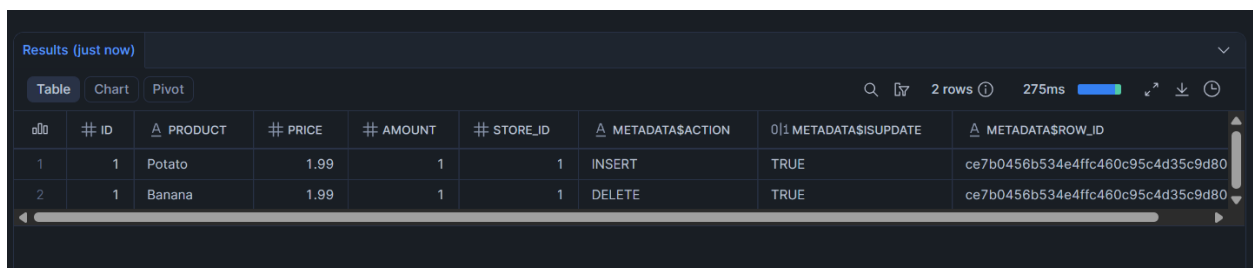
```
SELECT * FROM SALES_STREAM;
```

QUESTION : How does Snowflake represent an UPDATE operation inside a stream?

ANSWER : Snowflake represents an UPDATE as:

one DELETE record

one INSERT record inside the stream.



Results (just now)

Table Chart Pivot

2 rows 275ms

	# ID	PRODUCT	# PRICE	# AMOUNT	# STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
1	1	Potato	1.99	1	1	INSERT	TRUE	ce7b0456b534e4ffc460c95c4d35c9d80
2	1	Banana	1.99	1	1	DELETE	TRUE	ce7b0456b534e4ffc460c95c4d35c9d80

TASK 22 : Consume Update Using MERGE

```
MERGE INTO SALES_FINAL_TABLE tgt

USING (

    SELECT *

    FROM SALES_STREAM

    WHERE METADATA$ACTION = 'INSERT'

    AND METADATA$ISUPDATE = TRUE

) src

ON tgt.id = src.id

WHEN MATCHED THEN

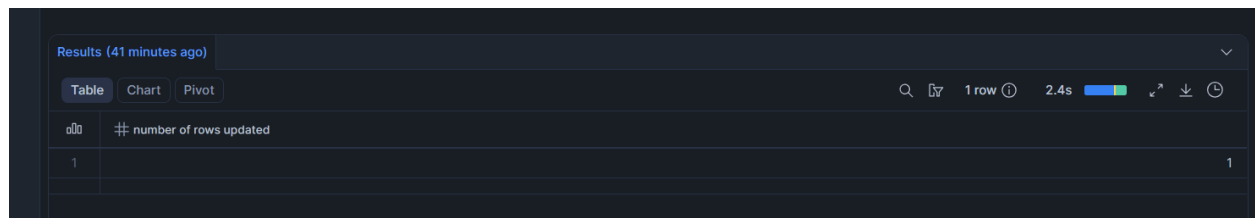
UPDATE SET

    tgt.product = src.product,

    tgt.price = src.price,

    tgt.amount = src.amount,

    tgt.store_id = src.store_id;
```



The screenshot shows a database query results window. At the top, it says "Results (41 minutes ago)". Below this, there are tabs for "Table", "Chart", and "Pivot". To the right of the tabs, there are icons for search, filters, and a summary: "1 row", "2.4s", and a progress bar. The main area displays a table with one row. The first column is labeled "number of rows updated" and contains the value "1".

number of rows updated
1

TASK 23 : Verify First Update

SELECT * FROM SALES_FINAL_TABLE;

SELECT * FROM SALES_RAW_STAGING;

SELECT * FROM SALES_STREAM;

QUESTION : Is Banana changed to Potato in SALES_FINAL_TABLE?

ANSWER : Yes. Banana is changed to Potato in SALES_FINAL_TABLE.

Results (just now)

Table Chart Pivot

9 rows 1.8s

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Potato	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	
9	9	Tomato	3.99	1	2	London	12

Results (just now)

Table Chart Pivot

9 rows 77ms

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Potato	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1
8	8	Paprika	4.99	1	
9	9	Tomato	3.99	1	2

Results (just now)

Table Chart Pivot

0 rows 776ms

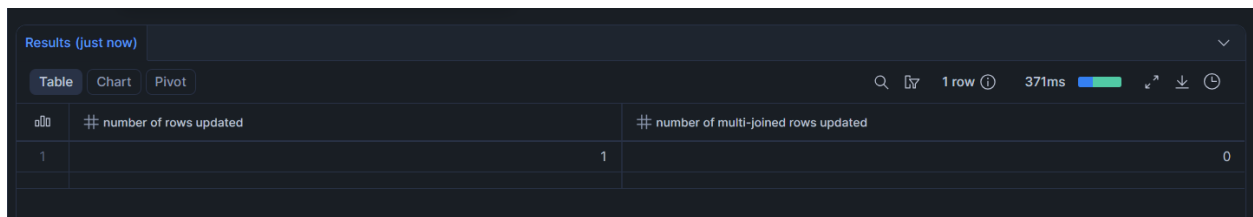
ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
----	---------	-------	--------	----------	------------------	--------------------	------------------

TASK 24 : Perform Second Update

```
UPDATE SALES_RAW_STAGING
```

```
SET product = 'Green apple'
```

```
WHERE product = 'Apple';
```



	# number of rows updated	# number of multi-joined rows updated
1	1	0

TASK 25 : Consume Second Update Using MERGE

```
MERGE INTO SALES_FINAL_TABLE tgt
```

```
USING (
```

```
SELECT *
```

```
FROM SALES_STREAM
```

```
WHERE METADATA$ACTION = 'INSERT'
```

```
AND METADATA$ISUPDATE = TRUE
```

```
) src
```

```
ON tgt.id = src.id
```

```
WHEN MATCHED THEN
```

```
UPDATE SET
```

```
tgt.product = src.product,
```

```
tgt.price = src.price,
```

```
tgt.amount = src.amount,
```

```
tgt.store_id = src.store_id;
```

Results (just now)	
Table	Chart Pivot
1 row	1.5s
# number of rows updated	
1	1

TASK 26 : Verify Final Output

SELECT * FROM SALES_FINAL_TABLE;

SELECT * FROM SALES_RAW_STAGING;

SELECT * FROM SALES_STREAM;

Results (just now)

TableChartPivot

🔍🔗9 rows ⓘ97ms

↕️⬇️🕒

#	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Potato	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Green apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	
9	9	Tomato	3.99	1	2	London	12

Results (just now)

TableChartPivot

🔍🔗9 rows95ms

	# ID	PRODUCT	# PRICE	# AMOUNT	# STORE_ID
1	1	Potato	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Green apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1
8	8	Paprika	4.99	1	
9	9	Tomato	3.99	1	2

Results (just now)

TableChartPivot

0 rows

186ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
--	----	---------	-------	--------	----------	------------------	--------------------	------------------

QUESTION : Is Apple changed to Green apple in SALES_FINAL_TABLE?

ANSWER : Yes.

Apple is changed to Green apple in SALES_FINAL_TABLE.